

Data Copilot: Client Onboarding Report

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KPIs

Total Revenue

13591643.7

Total revenue is a key indicator of business performance, reflecting overall sales success.

Average Order Value

137.75

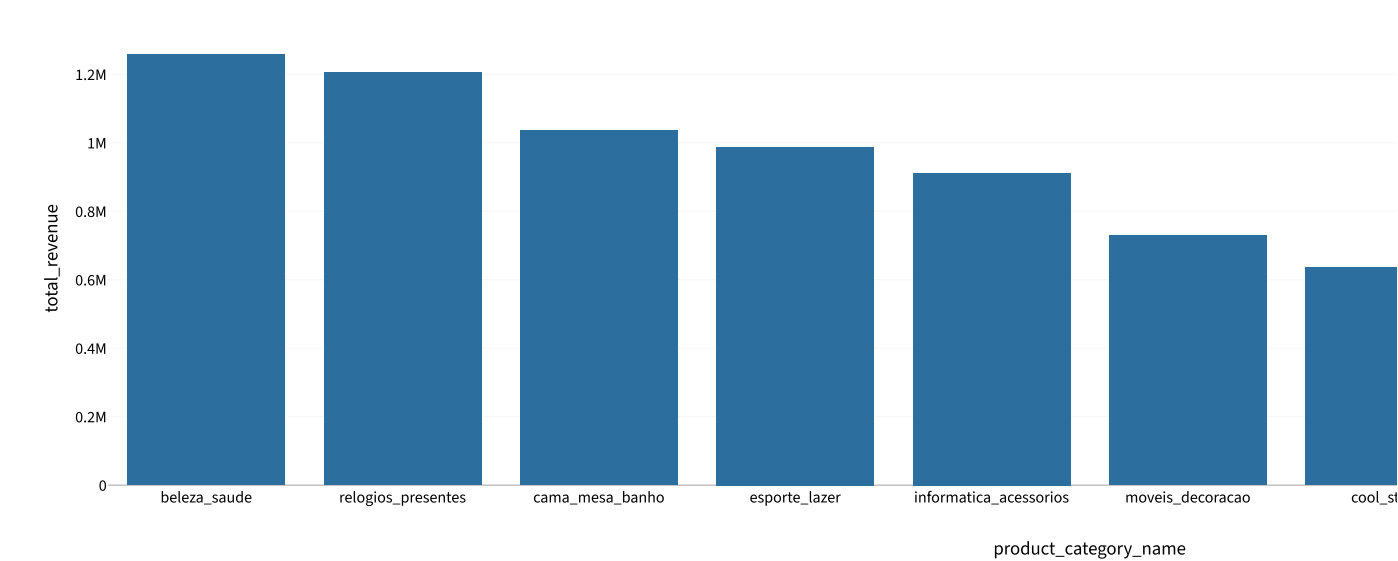
Average order value helps assess customer spending behavior and can guide pricing strategies.

Average Review Score

4.09

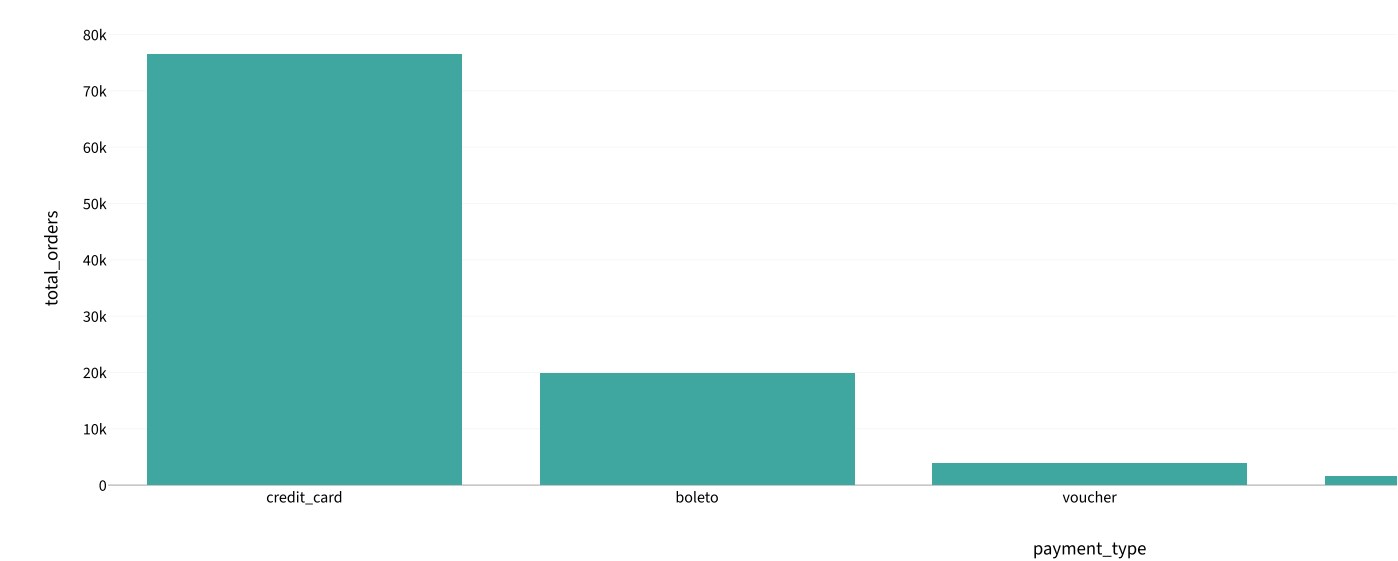
Average review score is crucial for assessing customer satisfaction and product quality.

Revenue by Product Category



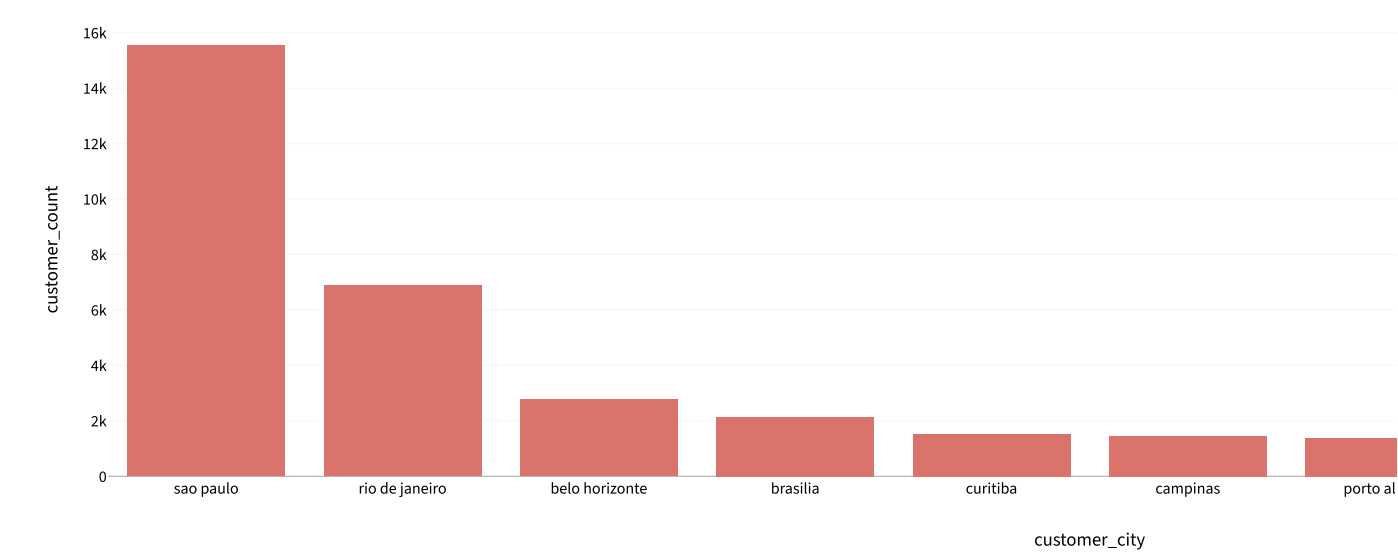
Understanding revenue by product category helps identify top-performing categories and informs inventory and marketing strategies.

Orders by Payment Type



Analyzing orders by payment type reveals customer preferences and can influence payment processing strategies.

Top 10 Cities by Customer Count



Identifying top cities by customer count can help target marketing efforts and optimize logistics.

Business Insights

- Total revenue for the period is approximately \$13.6 million, indicating strong sales performance and overall business health.
- The average order value is around \$137.75, which suggests that customers are spending a healthy amount per transaction; this insight can help refine pricing strategies.
- The top product category is 'beleza_saude' with over \$1.25 million in revenue, highlighting a key area for inventory focus and marketing efforts.
- Most orders are paid for using credit cards, accounting for 76,505 transactions; understanding this preference can enhance payment processing strategies.
- The average review score is 4.09, reflecting generally high customer satisfaction, but the high percentage of null values in review comments (88.34%) raises concerns about the completeness of feedback, potentially impacting product improvement efforts.
- There are 261,831 duplicate rows in the geolocation data, which could lead to inaccuracies in customer location insights, affecting targeted marketing and logistics planning.

What Else This Data Could Support

- Churn prediction could be implemented by analyzing customer purchase patterns and review scores.
- Demand forecasting can be enhanced by utilizing historical sales data from the 'order_items' table along with seasonal trends.
- Customer segmentation analysis could be conducted to tailor marketing strategies based on purchasing behavior and demographics.

▼ Technical details (data model, quality checks)

Data Quality Findings

- The 'geolocation' table has 261831 duplicate rows, which may lead to inaccurate geolocation data.
- The 'order_reviews' table has a high percentage of null values (88.34%) in the 'review_comment_title' column, indicating potential data collection issues.
- The 'order_payments' table has outliers in 'payment_installments' (6313 distinct values), which may skew financial analysis.

Checks performed, table by table

Table	Rows	Columns checked	Duplicate rows	Columns with missing values	Worst missing value c
customers	99441	5	0	0	none
geolocation	1000163	5	261831	0	none
order_items	112650	7	0	0	none
order_payments	103886	5	0	0	none
order_reviews	99224	7	0	2	review_comment_titl
orders	99441	8	0	3	order_delivered_cust
products	32951	9	0	8	product_category_na
sellers	3095	4	0	0	none
product_category_name_tra	71	2	0	0	none

Relationships confirmed between tables (checked by real matching values, not just column names):

- customers.customer_id → orders.customer_id (100.0% of values matched)
- order_items.order_id → orders.order_id (100.0% of values matched)
- order_items.product_id → products.product_id (100.0% of values matched)
- order_items.seller_id → sellers.seller_id (100.0% of values matched)
- order_payments.order_id → orders.order_id (100.0% of values matched)
- order_reviews.order_id → orders.order_id (100.0% of values matched)
- orders.customer_id → customers.customer_id (100.0% of values matched)

Recommended Data Model

Fact table: order_items

Dimensions: customers, products, sellers, order_payments, order_reviews

The 'order_items' table serves as the central fact table as it contains transactional data related to each order, including price and freight value. The dimension tables provide context such as customer details, product information, seller data, payment methods, and customer reviews, enabling comprehensive analysis of sales performance.

Entity-relationship diagram, generated from the actual confirmed keys:

